Quantium Virtual Internship

Retail Strategy & Analytics
Task 1 — Customer Analytics Report

Prepared for:	Julia – Category Manager, Chips
Prepared by:	Quantium Analytics Team
Analysis Period:	1 July 2018 – 30 June 2019
Dataset:	264,836 transactions | 72,637 customers

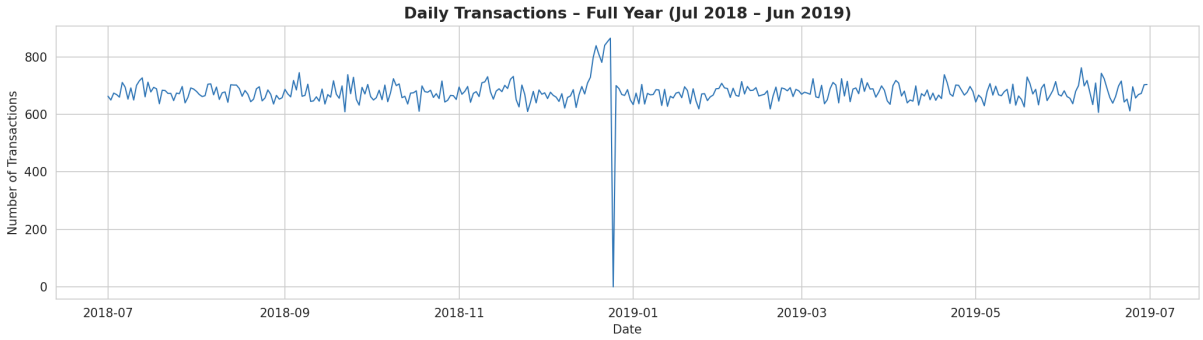
1. Data Preparation & Cleaning

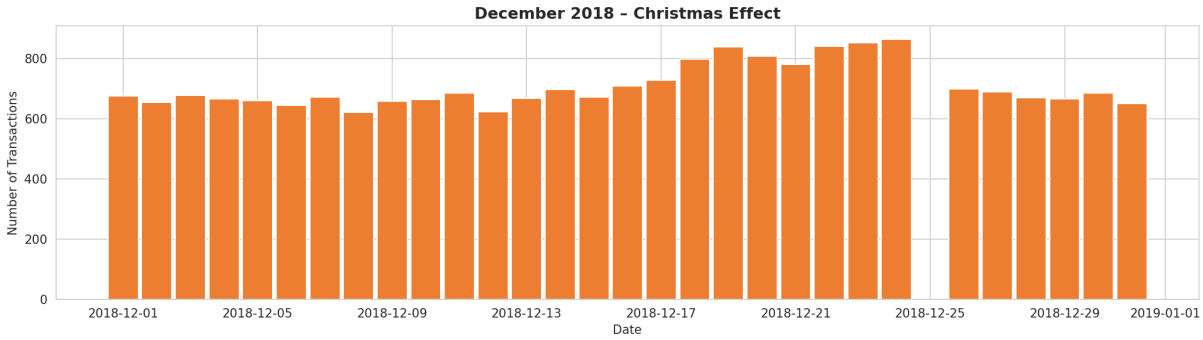
Before analysis, the raw transaction and customer datasets were thoroughly examined and cleaned. Key steps are summarised below.

1.1 Transaction Data

Step	Finding	Action Taken
Date Format	Stored as Excel integer	Converted to proper date (origin: 30 Dec 1899)
Salsa Products	18,094 non-chip rows found	Removed – out of scope for chip category
Outlier Customer	Card #226000 bought 200 packs twice	Removed – likely commercial buyer
Pack Size	Extracted from product name	New column PACK_SIZE (70g – 380g)
Brand Name	Extracted from first word	New column BRAND; aliases standardised
Missing Date	25 Dec 2018 had zero transactions	Expected – shops closed on Christmas Day

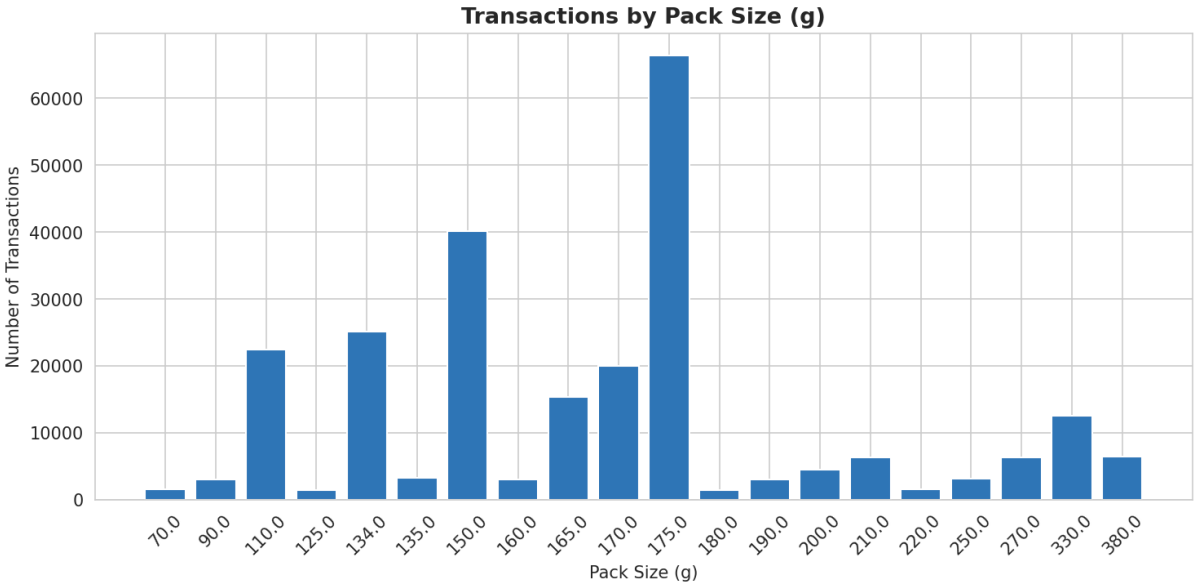
1.2 Transactions Over Time





■ **Insight:** A strong spike in chip purchases occurs in the lead-up to Christmas (Dec 15–24). Zero sales on 25 Dec are consistent with store closures. This seasonal pattern should inform promotional scheduling for H2.

1.3 Pack Size Distribution

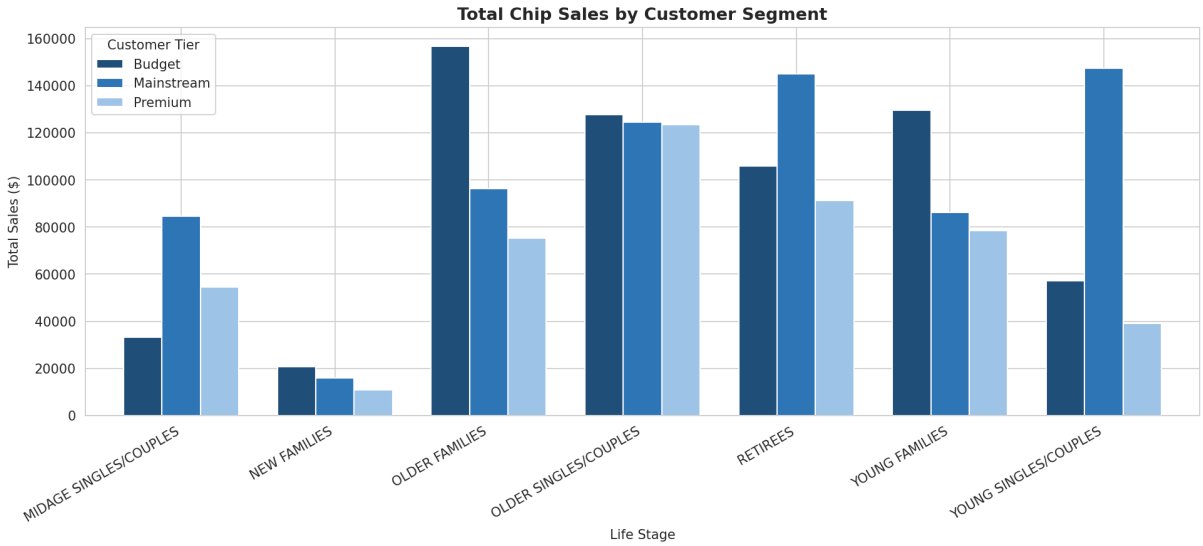


■ **Insight:** The 175g pack is overwhelmingly the most popular, followed by 150g and 110g. This suggests customers prefer mid-to-large sharing packs.

2. Customer Segment Analysis

After merging the cleaned transaction data with customer profiles (left join on LYLTY_CARD_NBR), we produced a dataset of 246,740 transactions with zero unmatched customers. We then defined four key metrics to evaluate each segment.

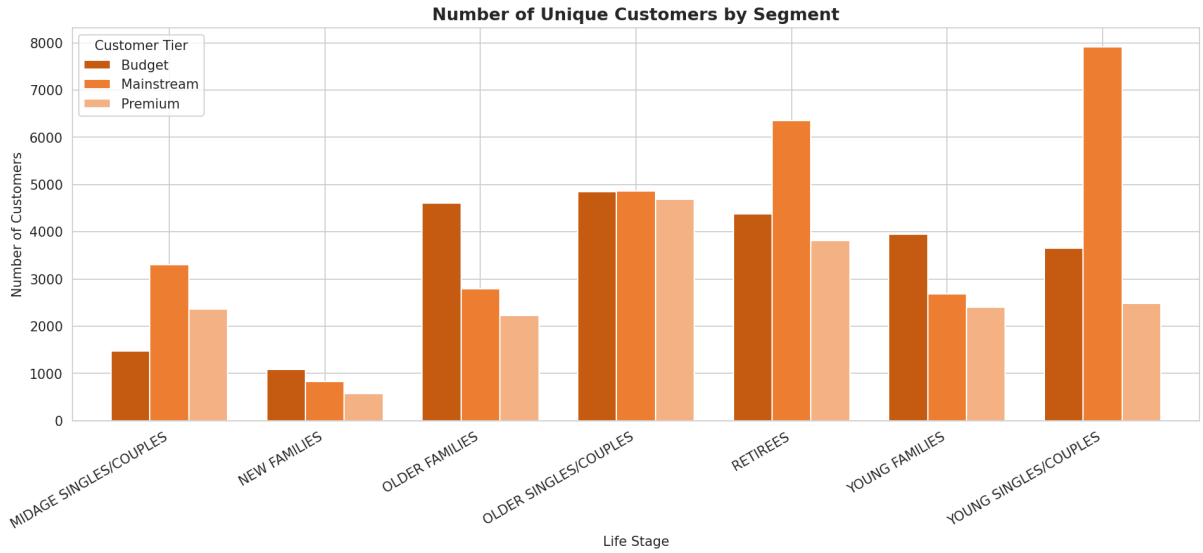
2.1 Total Sales by Segment



Rank	Life Stage	Customer Tier	Total Sales ($)
1	Older Families	Budget	$156,864
2	Young Singles/Couples	Mainstream	$147,582
3	Retirees	Mainstream	$145,169
4	Young Families	Budget	$129,718
5	Older Singles/Couples	Budget	$127,834

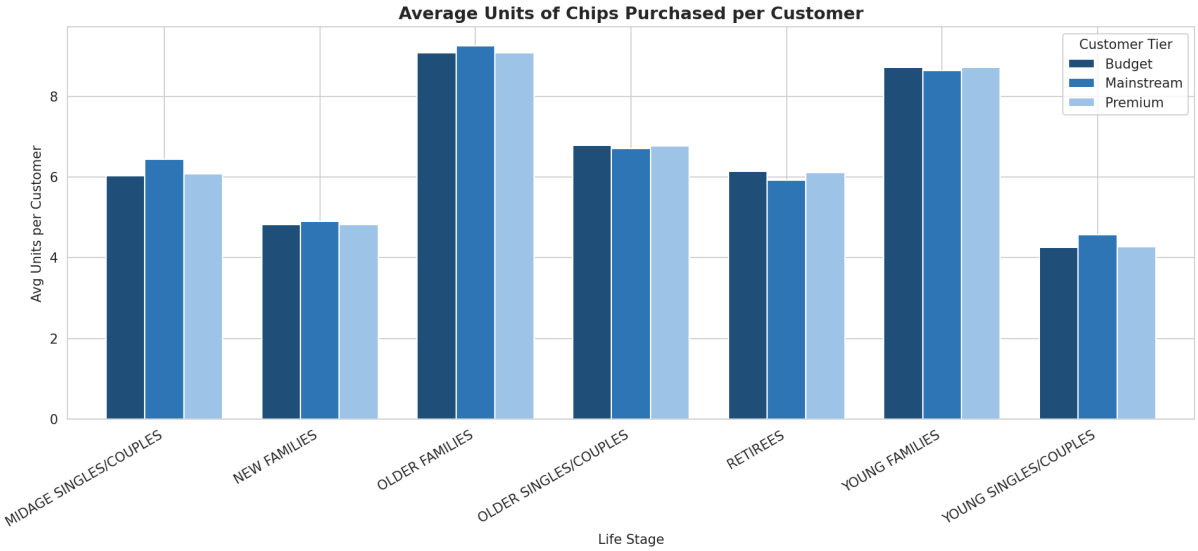
■ **Insight:** Budget – Older Families is the top revenue segment. Mainstream – Young Singles/Couples and Mainstream – Retirees are strong runners-up.

2.2 Number of Customers per Segment



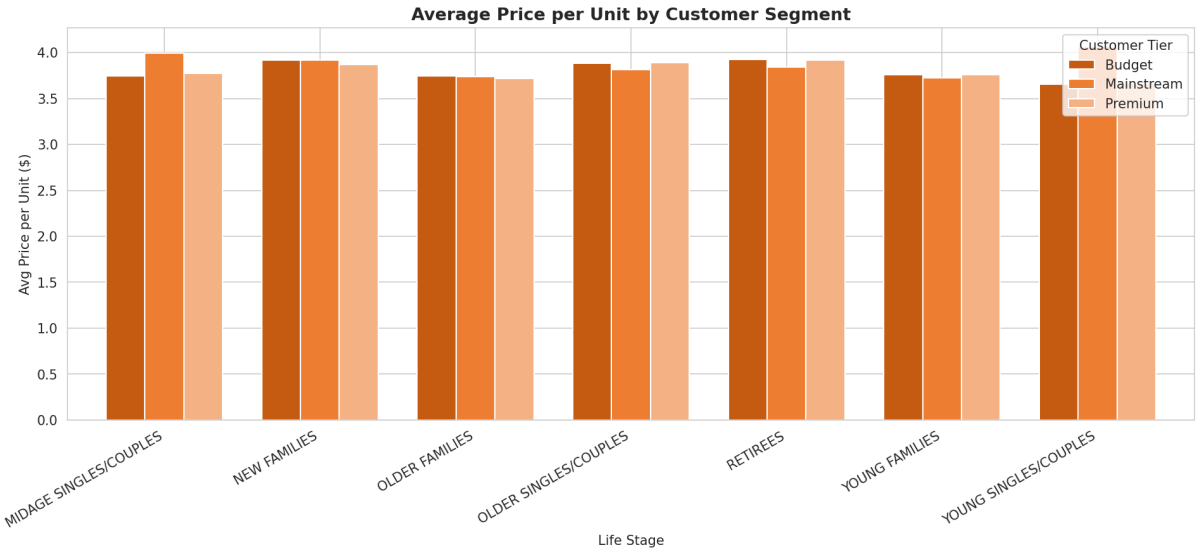
■ **Insight:** Mainstream Young Singles/Couples and Mainstream Retirees have the highest number of unique buyers. This volume drives their strong revenue figures — but Budget Older Families achieve top sales with fewer customers, suggesting higher spend per head.

2.3 Average Units Purchased per Customer



■ **Insight:** Older Families and Young Families buy significantly more chip units per customer than other segments. Families tend to buy in bulk — likely for household consumption and children's snacking.

2.4 Average Price per Unit



■ **Insight:** Mainstream Young and Mid-age Singles/Couples pay slightly more per unit than their Budget and Premium counterparts. Premium customers may spend their snack budget on healthier alternatives rather than chips.

2.5 Statistical Test – Price Differences

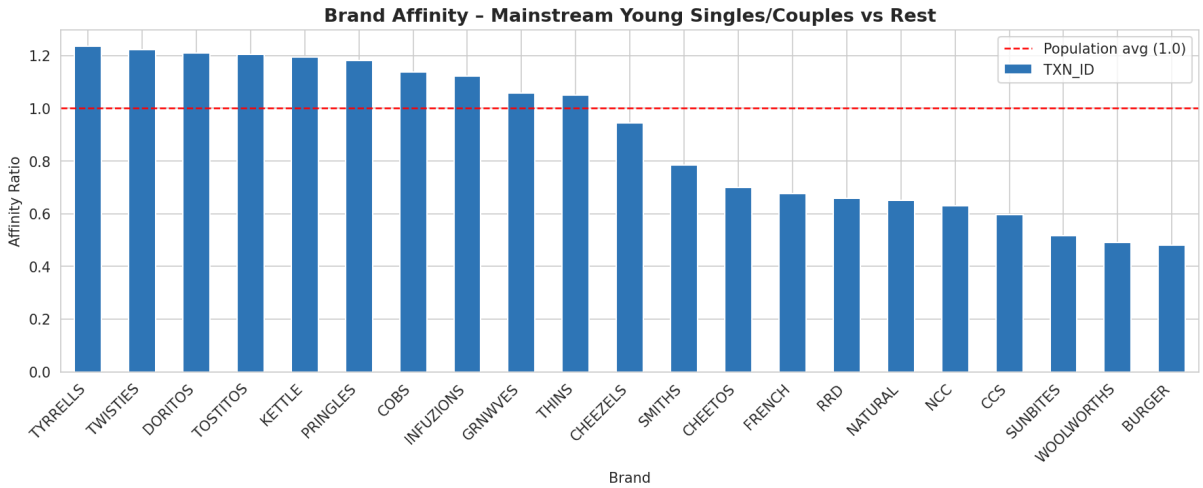
An independent samples t-test was conducted to determine whether the higher average unit price paid by Mainstream Young and Mid-age Singles/Couples is statistically meaningful.

Test	t-statistic	p-value	Conclusion
Mainstream vs Budget+Premium (Young & Midage Singles/Couples)	37.83	< 0.000001	SIGNIFICANT — Mainstream segment pays statistically more pe

3. Deep Dive — Mainstream Young Singles/Couples

This segment is our second-largest by revenue and has the highest customer volume. We analysed brand and pack size preferences compared to the rest of the population using an affinity ratio (segment share ÷ population share). A ratio above 1.0 means the segment over-indexes on that product.

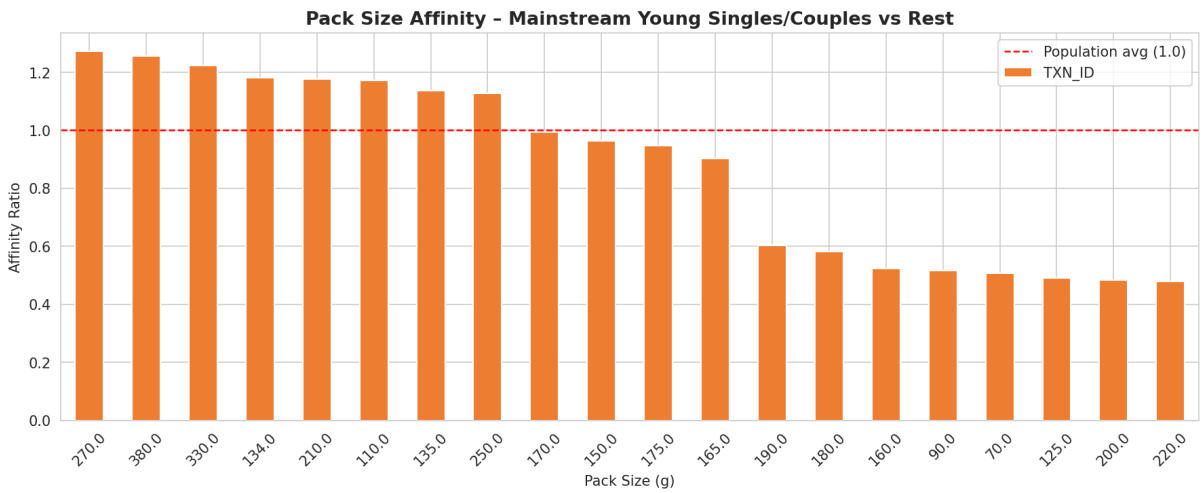
3.1 Brand Affinity



Brand	Affinity Ratio	Interpretation
Tyrrells	1.24	24% more likely to buy than the average customer
Twisties	1.22	22% more likely to buy
Doritos	1.21	21% more likely to buy
Tostitos	1.21	21% more likely to buy
Kettle	1.19	19% more likely to buy

■ **Insight:** Mainstream Young Singles/Couples show a clear preference for premium/trendy brands (Tyrrells, Kettle) and flavourful party-style chips (Twisties, Doritos, Tostitos). These brands should be prominently placed and promoted to capture this segment.

3.2 Pack Size Affinity



Pack Size	Affinity Ratio	Interpretation
270g	1.27	27% more likely to buy than average
380g	1.26	26% more likely to buy

330g	1.22	22% more likely to buy
134g	1.18	18% more likely to buy
210g	1.18	18% more likely to buy

■ **Insight:** Young Singles/Couples prefer larger sharing packs (270g–380g). Despite living alone or as couples, they are not buying smaller snack packs — suggesting social/party-occasion purchasing behaviour.

4. Strategic Recommendations for Julia

■ Priority Segment 1: Budget – Older Families

Top revenue segment. They buy frequently and in large volumes. Recommend multi-buy promotions (e.g. buy 2 get 1 free) and larger value packs to increase basket size.

■ Priority Segment 2: Mainstream – Young Singles/Couples

High customer volume with a preference for premium brands (Tyrrells, Kettle) and large sharing packs. Recommend end-of-aisle placements, social media promotions, and stocking 270g–380g sizes prominently for this group.

■ Priority Segment 3: Mainstream – Retirees

Third largest revenue contributor with a high customer count. Loyalty promotions and classic/familiar chip brands will resonate with this group.

■ Pack Size Strategy

Stock 175g as the core SKU (most popular overall). For Mainstream Young Singles/Couples, ensure strong availability of 270g–380g packs — these are significantly over-indexed.

■■ Pricing

Mainstream customers pay statistically more per unit ($p < 0.0001$). There is room to position premium brands at a slight price premium to this segment without reducing purchase intent.

■ Seasonal Planning

Sales spike sharply before Christmas. Ensure sufficient stock and run targeted promotions in the first two weeks of December.